DSAOB - NATURAL LANGUAGE PROCESSING

CONTINUOUS INTERNAL ASSESSMENT - 1

3/11/26

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01. (i) strict : 05/03/2025.

Regular Expression:

Given;

L1 : Appointment on 05/03/2025 at 10:30.

L2: call 9876543210 or +91-9123456789.

L3: Invalid : 5876543210 and 9876543210.

L4: Date 5/3/25 is not valid; mail ravi.k@college.edu.in.

(i) Requirements:

* It should be in the format of dd/mm/yyyy.
* date should be 01-31
* Month : 1-12
* year should be 4 digits.

Regular Expression:

\b(0[1-9][12][0-9]|3[01])/(0[1-9]|1[0-2])/[0-9]{4}\b.

Explanation:

| pattern | Meaning |
|---|---|
| 0[1-9] | 0-9 days. |
| [12][0-9] | 10-20 days. |
| 3[01] | 30-31 days. |
| 0[1-9] | 1-9 months |
| 1[0-2] | 10-12 months. |
| [0-9]{4} | Exactly 4 digit year. |

valid : 05/03/2025

Invalid: 5/3/25.

(ii) Regular expression for phone number:

Requirements:

* +91 = optional prefix followed by hyphen or space.
* OR leading zero.
* Number should start between 6 and 9.
* +91 should be Indian country code.
* Exactly 10 digits.

Regular Expression:

(?!<\d)(?:?:+91[-]|?:[0])?[6-9][0-9]{9}(?<\d)

(?!<\d) - NO digits immediatly before the number.

(?:?:+91[-]) - Indian country code.

[0] - optional leading zero.

[6-9][0-9] - Number starting with 6-9.

{9} - remaining 9 digits.

(?<\d) - No digits immediatly After the number.

(iii) Regular Expression for E-mail address:

Requirements:

* Username
* Mandatory @ symbol.
* domine name.
* date.
* specific name ends with atleast 2 letters.

Regular Expression:

[A-za-z0-9-%.A-]+@[A-za-z0-9-]+\[A-za-z]{2,}.

[A-2a-zo-9-%.4-] - user name.

@ - @ symbol.

[A-za-zo-9] - domine name.

[A-za-z] {2,} -

(iv) False Method:

Given:

5876543210 and 98765432101.

Example:

⇒ 5876543210

the given number is invalid as it starts with 5.

Because the phone number should always start between 6 - 9. so it's Invalid.

⇒ 98765432101:

the given number is invalid as it is in 11 digits.

The phone number should always be in 10 digits number.

Date Substitution:

plain text : 05/05/2025

Required input : 2025-05-05

Find pattern :

(\d{2})/(\d{2})/(\d{4})

$3-$2-$1

Python Code:

```
import re
text = "Appointment on 05/05/2025 at 10:30"
```

```
result = re.sub(
        r'(\d{2})/(\d{2})/(\d{4})',
        r'\g<3>-\g<2>-\g<1>',
        text
)
print(result)
```

Output: "Appointment on 2025-03-05 at 10:30".

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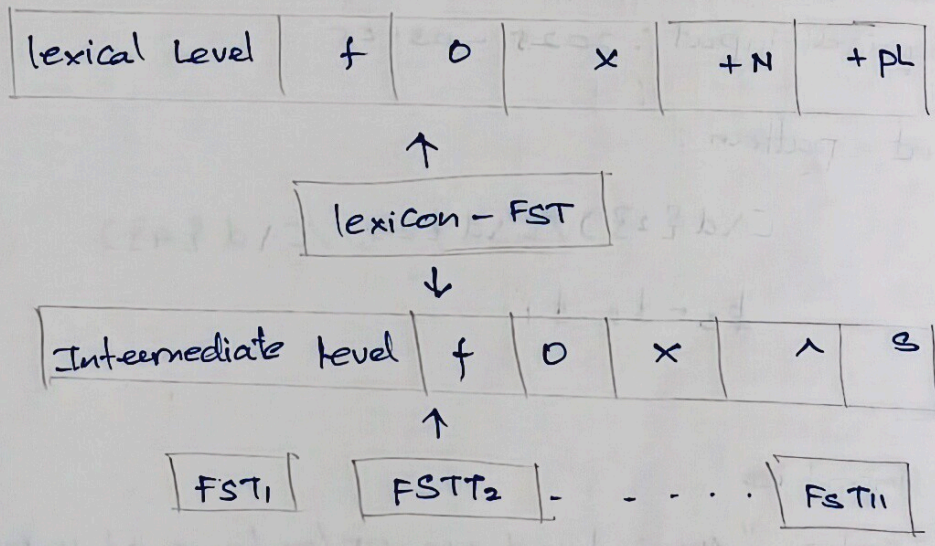
02. Finite-state Morphological Analysis:

(a) Finite-state automaton for English plural formation:

A finite-state morphological analyzer consists of three levels.

* lexical level : Root + grammatical morpheme.
* Intermediate level: Morphological representation.
* Surface level: Actual written word.

Morphological Analysis:



(b) Morpheme sequence:

| Word | Morpheme | Sequence. |
|---|---|---|
| dogs | dog + s | yes |
| mice | mouse + s | yes with changes |
| geese | goose + es | yes with changes |
| foxes | fox + es | yes. |

(c) lexical, Intermidiate and surface form:

| lexical form | Intermidiate form | surface form. |
|---|---|---|
| dog + s | dog + s | dogs |
| mouse + s | mouse + s | mice |
| goose + es | goose + es | geese |
| fox + es | fox + es | foxes. |

(d) Inflectional vs derivational morphology:

| Word | Types | Justification. |
|---|---|---|
| Walked | Inflectional | Past tense of walk |
| unhappiness | Derivational | Create new word from happy. |
| Computerization | Derivational | Create new word from computerise. |
| Geese | Inflectional. | Irreg plural of goose. |

3. (i) Porter Stemmer:

It reduces the word by using suffix removal rules.

word given:

ponies, monitoring, relational, hopefulness.

and generalizations.

| Word | Step | Roll | Result |
|---|---|---|---|
| ponies | 1a | ies-i | Poni |
| Monitoring | 1b | ing-removed | Monitor |
| relational | 2 | ational-ate | Relate |
| hopefulness | 3 | fulness-ful | hopeful |
| generalizations. | 1a | s-removed | generalization. |

(ii) A training Corpus has three sentence:

<s> the cat rat </s> = s1

<s> the rat ran <s1> = ~~s3~~

<s> the dog sat <s /> = s2

s3 = the dog ran.

unsmoothed Bigram:

$$P(w_i \mid w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

s1 = " the cat sat "

$$P(\text{the} \mid <s>) = \frac{3}{3} = 1$$

$$P(\text{cat} \mid \text{the}) = \frac{2}{3}$$

$$P(\text{sat} \mid \text{cat}) = \frac{1}{2}$$

$P(</s> | sat) = \frac{2}{2} = 1$

$P(S_1) = 1 \times \frac{2}{3} \times \frac{1}{2} \times 1 = \frac{2}{6} = \frac{1}{3}$

$P(S_1) = \frac{1}{3} = 0.333.$

$S_2 =$ "the dog sat"

$P(the | <s>) = \frac{3}{3} = 1$

$P(dog | the) = \frac{1}{3}$

$P(sat | dog) = \frac{1}{2}$

$P(</s> | sat) = \frac{2}{2} = 1$

$P(S_2) = 1 \times \frac{1}{3} \times \frac{1}{2} \times 1 = \frac{1}{6}$

$P(S_2) = 0.166.$

05. (a) Viterbi algorithm and best tag sequence:

$V_1(PRP) = 0.5 \times 0.8 = 0.4$

$V_2$ is for "can" and $V_3$ is for "fish"

MD $0.4 \times 0.4 \times 0.3 = 0.048$

VB $0.4 \times 0.5 \times 0.02 = 0.004 \max(0.048 \times 0.8, 0.04 \times 0.1, 0.0004 \times 0.4) \times 0.01 = 0.000384.$

Now for:

NN $\Rightarrow 0.4 \times 0.1 \times 0.01 = 0.0004 \max(0.048 \times 0.2, 0.004 \times 0.5, 0.0004 \times 0.5) 0.02 = 0.000192.$

Among these, the best final state is VB. The back-tracing they/PRP can/MD fish/VB. all there with the probability $3.84 \times 10^{-4}$. And then half of the probability is competing reading of they/PRP can/MD fish/NN.

(b) Rule - Based :

These are two rules in this , and they are.

Rule 1 : If the word is "can" with the tags (MD, VB, NN) and previous word is personal pronoun and it eliminates NN and VB , so then can = MD.

Rule 2:

If the previous tag was MD, eliminate NN . So then fish = VB. Contrated rules automatically eliminates readings from the set until one last tag is left.

(c) Transformation Based:

they / PRP can / MD fish / NN, all has 1 error out of the 3 tokens. The learner tries the templates over the previous tag context and the labels the best one. they / PRP can / MD fish / NN. And then the score = 46 - 6. = 40.

(d) Comparison :

Rule - Based Strengths and its weakness :

Rule - Based hand written constraints and lexican. Costly to write , . Transparent and also hard to maintains. Stochastic tags and words . probability from words and learns automatically from the words and its too hard.

04. (a)

| Unigrams | | Bigrams. |
|---|---|---|
| <s> | 3 | (<s> I) = 3 |
| I | 3 | I like = 2 |
| like | 2 | I drink = 1 |
| tea | 2 | like tea = 1 |
| coffee | 1 | like coffee = 1 |
| drink | 1 | drink tea = 1 |
| </s> | 3 | tea </s> = 2 |
| | | coffee </s> = 1 |

$$P(\text{Tea} / \text{like}) = \frac{1}{2} = 0.5$$

$$P(\text{drink} / \text{like}) = \frac{0}{2} = 0.$$

(b)

$$P(N/V) = \frac{C(V,W) + 1}{}$$